

Title - Developing a business intelligence system for decision support for a real-life business - Walmart

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Company Introduction

Walmart is the biggest player in the global retail chain and has been the topmost company by revenue for more than a decade. It was founded by Sam Walton in Arkansas, US in 1962. The company has grown from a single discount store into a growing empire of over 10,623 stores across 24 countries (Walmart, 2023). It is home to about a total of 2.1 million associates worldwide. In 2023 alone, this retail giant generated upwards of \$611 billion (Walmart, 2023) in revenue which is a testament to its immense reach and enduring appeal.

Walmart is committed to its core mission of "EveryDay Low Price" and has consistently delivered the lowest prices to customers, restructuring the retail industry with its innovative approach. Its primary focus on customer service, respect for associates, striving for excellence, and acting ethically are deeply embedded in its core principles, rising to unparalleled excellence, establishing itself as a household name linked with consistent quality and affordability.

Apart from its sheer size and global reach, Walmart takes pride in several unique selling points. Its unique supply chain management system and cutting-edge technology, allows for efficient delivery of goods at low costs, ensuring its commitment to Everyday Low Prices. To keep the costs minimum throughout the supply chain, Walmart uses business intelligence systems that enables it to make informed decisions to optimise its business strategy.

By harnessing the power of data analytics, Walmart is able to gain key insights into customer behaviour, market trends, and operational efficiencies, enabling it to make informed decisions that improve its strategy and further enhance its business. Along its path towards growth and innovation, Walmart's commitment to BI remains a vital component of its success. By using this powerful tool, Walmart can ensure informed decision-making, optimise operations, and continue delivering its promise, solidifying its position as a retail powerhouse for the years ahead.

Problem Identification

Ageing stores & systems

Walmart faces a lot of challenges due to its vast operational scope. First among these is the weighty burden of outdated infrastructure engrossed within its extensive network of stores. There is a need for immediate renovation or overhaul of these old establishments that hinders Walmart's ability to create a contemporary and immersive shopping atmosphere. Outmoded store layouts and fixtures present formidable obstacles, limiting the company's adaptability to dynamic market shifts, evolving customer preferences, and the emergence of new technologies. These rigid structures reduce Walmart's agility in swiftly aligning product assortments and store layouts, obstructing positive efforts to elevate the overall shopping experience. Further, these challenges are the amplified maintenance costs incurred by these old establishments, further straining Walmart's financial resources and impeding investments in transformative innovations.

Decentralised decision making

In parallel, Walmart struggles with the ramifications of its decentralised decision-making model, granting individual store managers commendable autonomy (Pandey et al., 2021). While this environment is ripe for innovation, the decentralised approach inadvertently engenders a mosaic of inconsistent practices across the conglomerate. This variability manifests in different product assortments, inconsistent pricing models, and divergent customer service standards across stores. While promoting localised innovation, this decentralised modus operandi elevates the risk of inconsistent stocking and operational disparities. The lack of standardised practices poses the imminent threat of stockouts, potentially disappointing customers and eroding Walmart's esteemed brand reputation. Compounding this challenge is the absence of real-time data on store sales, customer behaviours, and inventory levels, slowing down decision-making processes. Lethargic restocking periods intensify the issue that may further undermine customer satisfaction, potentially risk Walmart's market standing.

Furthermore, not keeping up with its competitors in terms of using hybrid technology (a combination of open source and in-house) has led Walmart's inability to capture important data metrics critical for understanding customer behaviour and optimising store operations. These limitations hinder comprehensive foot traffic analysis, restricting Walmart to identify visitation patterns and allocate resources efficiently. The inability to measure and analyse dwell times per

customer or per aisle hampers strategic decisions related to product placement and store layout enhancements, impacting the overall shopping experience. Identifying high-value buyers, particularly those arriving by car and occupying parking spaces, remains challenging due to infrastructure limitations. The inability to effectively track parking occupancy as an indicator of high-value buyers hampers targeted marketing strategies and personalised engagement initiatives aimed at this segment, potentially missing valuable opportunities for customer engagement and retention.

Addressing these challenges demands a comprehensive approach involving infrastructure modernisation, technological advancements, and streamlined operational strategies. Overhauling infrastructure, streamlining decision-making processes, and embracing advanced data capturing technologies stand as pivotal steps in Walmart's journey towards fostering operational efficiency and elevating customer experiences across its extensive retail landscape.

Literature Review

The challenges faced by Walmart require an in-depth exploration of state-of-the-art solutions. Outdated infrastructure, a primary concern for Walmart's store network, has been extensively analysed in recent studies. (Chatzoglou et al., 2022) identified a significant correlation between store layouts and customer satisfaction, advocating for modernised spaces conducive to positive shopping experiences. Concurrently, (Kraus et al., 2022) identified the inflexibility of outdated infrastructure as a major barrier to adapting to dynamic market trends. Emerging technologies showcase promising solutions. Smart shelves integrated with RFID tags and sensors, as researched by (Choi et al., 2012), offer real-time inventory data and optimise product placement, addressing inventory management challenges. More research is done to evaluate how digital sign boards which are dynamic in nature can help to personalise shopping experience and enhance product discovery by (Li et al., 2023). Virtual reality and augmented reality technologies, as studied by (Khan et al., 2022), present immersive experiences that can elevate customer engagement and transform store layouts. These findings establish the pivotal role of technology in revitalising Walmart's infrastructure to meet modern retail demands.

Decentralised decision-making, while fostering innovation, introduces inconsistencies and inefficiencies. (Zalmanson, 2018) emphasised the importance of balancing autonomy with centralised oversight to ensure a consistent brand experience and operational excellence.

Similarly, (Hwang and Park, 2016) proposed utilising data analytics tools to empower store managers with insights for informed decision-making, thereby reducing inconsistencies and enhancing efficiency. Streamlining decision-making processes through advanced analytics tools can harmonise Walmart's diverse operational practices, promoting standardisation without stifling innovation.

Real-time data's significance in retail decision-making has been extensively researched. (Hashmi, 2016) showcased its impact on inventory management and customer satisfaction. Similar studies have shown that real-time data is crucial in identifying customer trends and optimising marketing campaigns. Sensor networks and camera analytics, as suggested by (Hussein and Banimelhem, 2023), can capture real-time data on customer behaviour and store operations. Leveraging machine learning algorithms, Walmart can analyse this data to predict customer demand, streamline inventory management, and drive personalised marketing strategies, thereby mitigating the issues arising from the lack of real-time data.

Hybrid technology adoption presents immense potential for innovation and agility. (Chatzoglou et al., 2022) emphasised its benefits in enhancing retail innovation. Their research suggests hybrid technology's adaptability to diverse business needs and its capacity to integrate data across systems. By embracing hybrid technology, Walmart can amalgamate diverse data channels like foot traffic analysis, parking occupancy sensors, and customer dwell time tracking. Harnessing this data will facilitate a comprehensive understanding of customer behaviour, allowing for optimised store layouts, personalised marketing campaigns, and an enriched shopping experience.

Business Intelligence Solutions by the Company

Existing data analytics system

Walmart currently uses a strong and comprehensive in-house BI solution, carefully crafted using open-source technologies. This system operates by collecting data from numerous touch points across the business landscape, including physical stores, vendor sites, supplier integrations, and e-commerce platforms. The integration strategy revolves around employing multiple expert data connectors which are good at merging diverse datasets from various sources. This combined data then undergoes a rigorous preparation phase before being ingested into a large-scale pipeline.

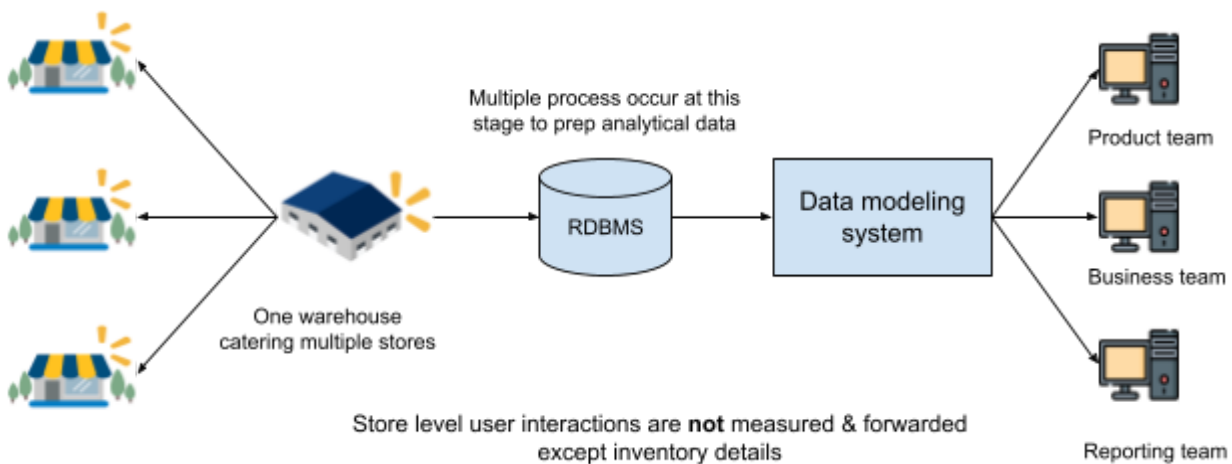


Fig 1: Current system at Walmart captures data from warehouse to forecast business needs

During the data preparation phase, a series of steps, including cleansing, sanitising, and outlier removal, are carefully executed to ensure data readiness for subsequent model training. Within a corporate giant like Walmart, a range of training methods, including both available market options and established techniques, are deployed. Some models are tailor-made to address specific business use-cases, and these models are integrated into a structured workflow alongside corresponding business.

Each model undergoes rigorous testing against dedicated training and test data sets, with periodic checks on accuracy and precision utilising a CI/CD (Continuous Integration and Continuous Deployment) approach. Upon receiving approval from product and business teams on modified trained models, they undergo deployment into production environments. Subsequently, these models begin a feedback loop, continuously gathering data for further enhancement and refinement.

This iterative cycle of model improvement, validation, deployment, and feedback forms the backbone of Walmart's BI solution, ensuring a dynamic and constantly evolving system geared towards optimisation and innovation in response to the complexities of their extensive business operations.



Fig 2: Internal steps of existing business intelligence system (Subramaniam, 2022)

The existing business intelligence (BI) model at Walmart can be characterised as a robust and integrated system that uses data-driven insights to drive decision-making and operational efficiency across the organisation.

Sample data set and prediction model

Fortunately, we came across a valuable sample dataset for Walmart stores (Kaggle, 2018). This dataset, consisting of more than 6,000 rows, has detailed information about various aspects of each sale, including store number, date, weekly sales amount, list of holidays, temperature, fuel price, and unemployment rate.

As step1 of the data refinement process, we converted the list of holidays into a boolean column. By using excel we were able to create a new column and mark it as “1” whenever the string was matching with the holiday name. (= IF(cell = ” holiday_name”, 1, 0)). Once all the values were numeral it was ideal for conducting predictive analysis through Rapid Miner.

Weekly_Sales Number	CPI Number	Fuel_Price Number	Holiday_Flag Number	Store Number	Temperature Number	Unemployment Number
1643690.900	211.096	2.572	0	1	42.310	8.106
1641957.440	211.242	2.548	1	1	38.510	8.106
1611968.170	211.289	2.514	0	1	39.930	8.106
1409727.590	211.320	2.561	0	1	46.630	8.106
1554806.680	211.350	2.625	0	1	46.500	8.106
1439541.590	211.381	2.667	0	1	57.790	8.106
1472515.790	211.216	2.720	0	1	54.580	8.106
1404429.920	211.018	2.732	0	1	51.450	8.106
1594968.280	210.820	2.719	0	1	62.270	7.808
1545418.530	210.623	2.770	0	1	65.860	7.808
1466058.280	210.489	2.808	0	1	66.320	7.808

Table 1: Sample data for Walmart weekly sales and other parameters

We employed two algorithms, General Linear Regression and Decision Tree, to predict weekly sales based on the available information.

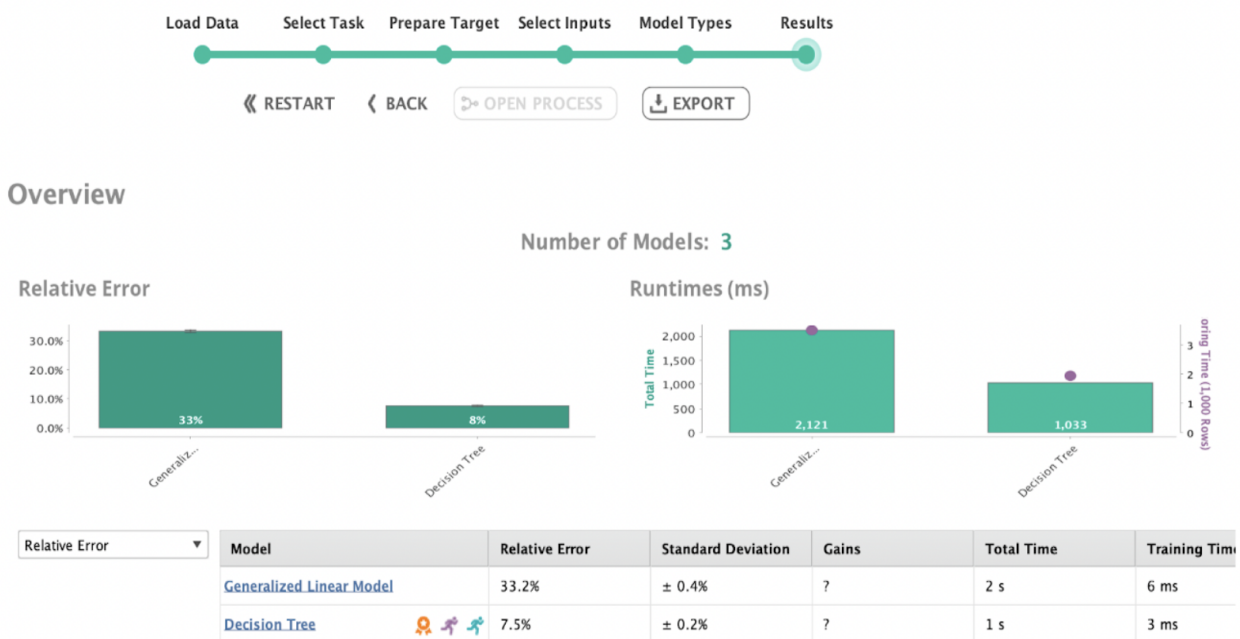


Fig 3: Relative error for both prediction model

While the General Linear Regression model offered a relative error of around 33%, the Decision Tree model performed significantly better, achieving an impressive 7.5% relative error. This suggests that the Decision Tree model effectively captured the complex relationships between the independent and dependent variables, leading to more accurate predictions. The results are further visualised in the provided images.

Generalized Linear Model – Predictions Chart

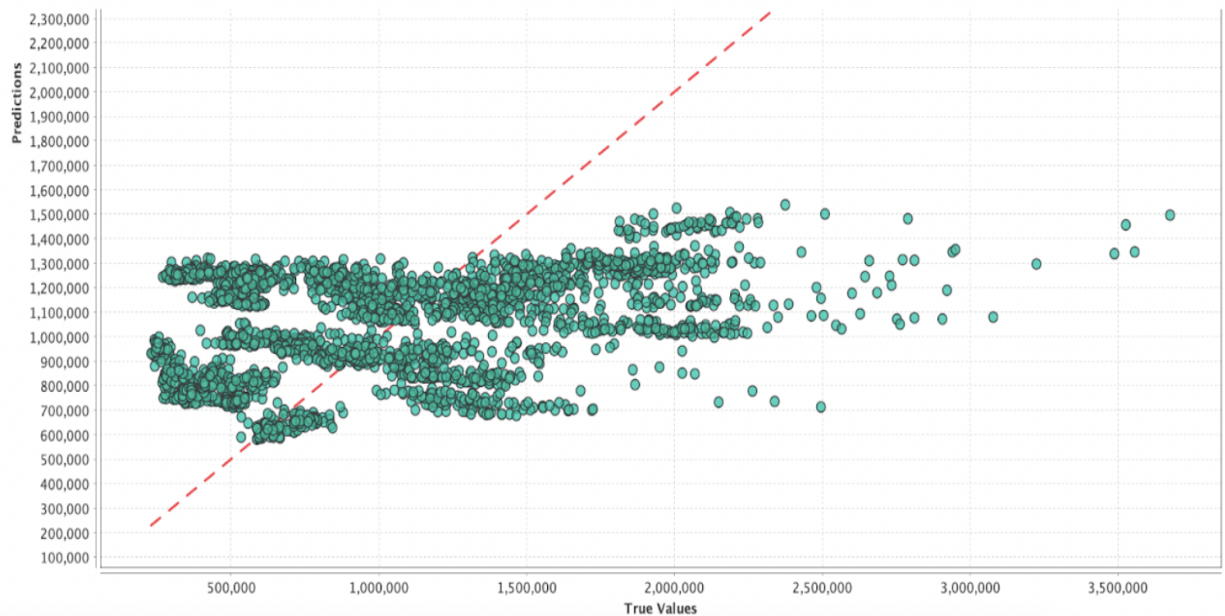


Fig 4: Most data points are not aligning with generalised linear model

Decision Tree – Predictions Chart



Fig 5: Most data points are aligning with decision tree prediction model

While we've achieved promising results with our weekly sales prediction model, we can further refine it by incorporating store-specific attributes related to user time spent. Analysing the number of aisles, product placement, signage effectiveness, interactive kiosks, staff interaction, event impact, WiFi tracking, camera analytics, and sensor data can provide valuable insights into customer behaviour and engagement. By leveraging these store-specific attributes, we can build

a more accurate user time prediction model, enabling us to optimise store layout, enhance the customer experience, and ultimately drive increased sales and profitability.

Proposing Solutions

The challenge of accurately capturing and leveraging customer interactions at the store involves a proactive strategy and that requires some infrastructural changes. Our suggestion is to convert existing brick and mortar stores to smart stores that are equipped with state of the art tools and technology to address the problem

At first we can integrate IoT sensors and smart technologies throughout stores, like Wi-Fi tracking and camera analytics. This facilitates real-time user movement analysis and interaction measurements. This investment in stores will enable a comprehensive understanding of customer behaviour, interactions with kiosks, and response to promotional displays.

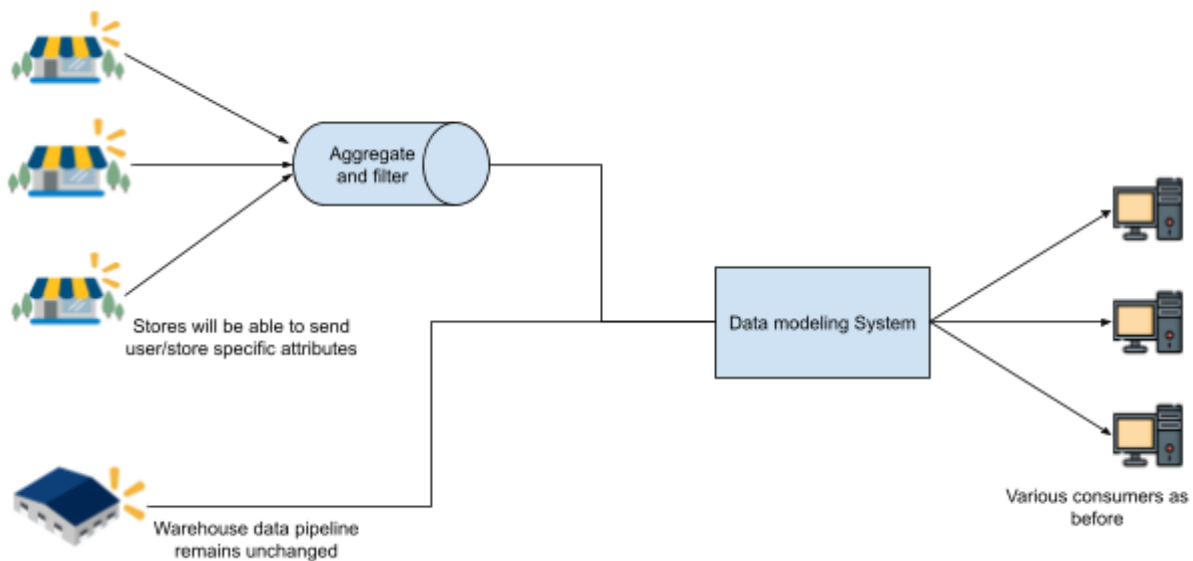


Fig 6: Proposed smart stores that can provide user engagement data to BI system

Secondly, we can employ a multi pipeline approach to push raw data into a data lake. This can be fine tuned to accommodate the diverse data essential across various business functions. This approach can be tailored to the unique needs of each dataset and functional area, ensuring streamlined operations and maximising the efficiency of data analysis. The concern regarding unstructured data and potential missing attributes, especially assortment values managed at the

store level, can impact the uniformity of the data fed into the data lake. However, using separate pipelines tailored for specific use-cases allows Walmart to navigate this challenge effectively.

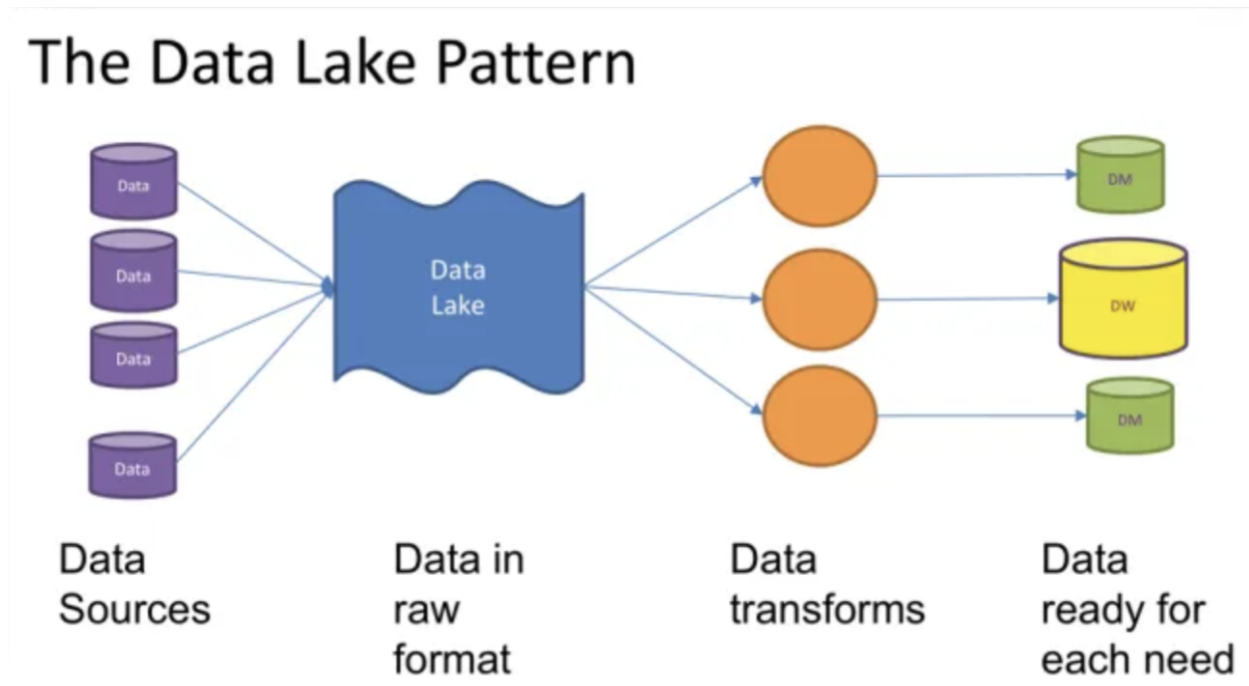


Fig 7: Integrate data lake pattern as some stores may have unstructured or missing attributes (datakitchen.io, 2019)

This segmentation enables the company to prioritise and process the data based on relevance and applicability. For use-cases where missing attributes, like assortment values, might interfere with analysis, Walmart can divert the data flow towards pipelines and analytics where such attributes are not crucial or necessary for insights.

Lastly, enhancing operational efficiency is necessary to get transformative changes in data analytics methodologies. Using real-time data capture and a multi-pipeline approach will help Walmart towards proactive decision-making and tailored customer experiences. The incorporation of advanced analytics derived from user time data facilitates accurate predictions of store dwell times, allowing optimisation of layout and product placements. Personalising customer interactions based on individual preferences and analysing engagement with specific products empowers targeted promotions, boosting customer engagement and satisfaction.

By accurately predicting user time spent in stores, we will be able to optimise our store layouts, product placements, and staffing schedules. These insights, derived from a deeper understanding of user behaviour, enable us to fine-tune the shopping experience, enhancing customer

satisfaction and engagement. Moreover, the ability to personalise promotions based on individual preferences and shopping patterns expand our revenue-driving strategies. Additionally, proactive inventory management, driven by real-time sales data, reduces stockouts and maximises sales opportunities.

Conclusion

Throughout this coursework, we have explored the multiple issues faced by Walmart. From the inefficiencies of decentralised decision-making to the outdated infrastructure and inventory management hurdles. We have identified issues demanding innovative resolutions. To address these challenges and help Walmart to move towards a new era of growth and success, we propose a data-driven transformation, fueled by cutting-edge technologies and a tailored multi-pipeline approach.

At the heart of our proposed solution lies the strategic deployment of real-time data capture technologies. By implementing sensors throughout stores, harnessing WiFi tracking capabilities, and leveraging the power of camera analytics, Walmart can gain an unprecedented understanding of customer behaviour and in-store dynamics. These insights, once captured and analysed, can be utilised to optimise store layouts, personalise customer experiences, and drive targeted promotions.

The true power of this data-driven approach lies in its prioritisation of critical data streams. By carefully analysing the impact of each data stream on immediate business needs and strategic goals, Walmart can ensure that resources are directed towards the most impactful areas. This prioritisation enables a more efficient allocation of resources, leading to streamlined operations and enhanced decision-making capabilities.

The practical implications of these data-driven solutions are very useful. We can imagine multiple possibilities where enhanced predictions of customer dwell time can guide store layout optimisation, leading to improved product discovery and increased sales. Personalised digital sign board and targeted promotions tailored to individual customer preferences can deliver a truly unique and engaging in-store experience, fostering customer loyalty and satisfaction.

The benefits extend far beyond the stores for example improved inventory management and accurate demand forecasts, can reduce stockouts and overstocking, optimising inventory costs and ensuring product availability. Also, with customer data Walmart can derive strategic decisions regarding marketing campaigns, product development, and pricing.

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